

8.FOURIER TRANSFORM AND FREQUENCY DOMAIN ANALYSIS

OBJECTIVE :

This experiment is divided into four simulations:

1. Fourier Transform of Periodic Impulse Train
2. Discrete Time Fourier Transform

Experiment 8.1

Fourier Transform of Periodic Impulse Train

Aim:

To compute and analyze the Fourier Transform of a periodic impulse train using MATLAB. The Aim is to observe the frequency-domain characteristics of a periodic impulse train, which should result in a discrete set of frequency components due to the periodicity of the signal.

Program:

```
% Fourier Transform of Periodic Impulse Train
```

```
clc;
```

```
clear;
```

```
close all;
```

```
% Time axis
```

```
t = -5:0.01:5;
```

```
% Generate periodic impulse train
```

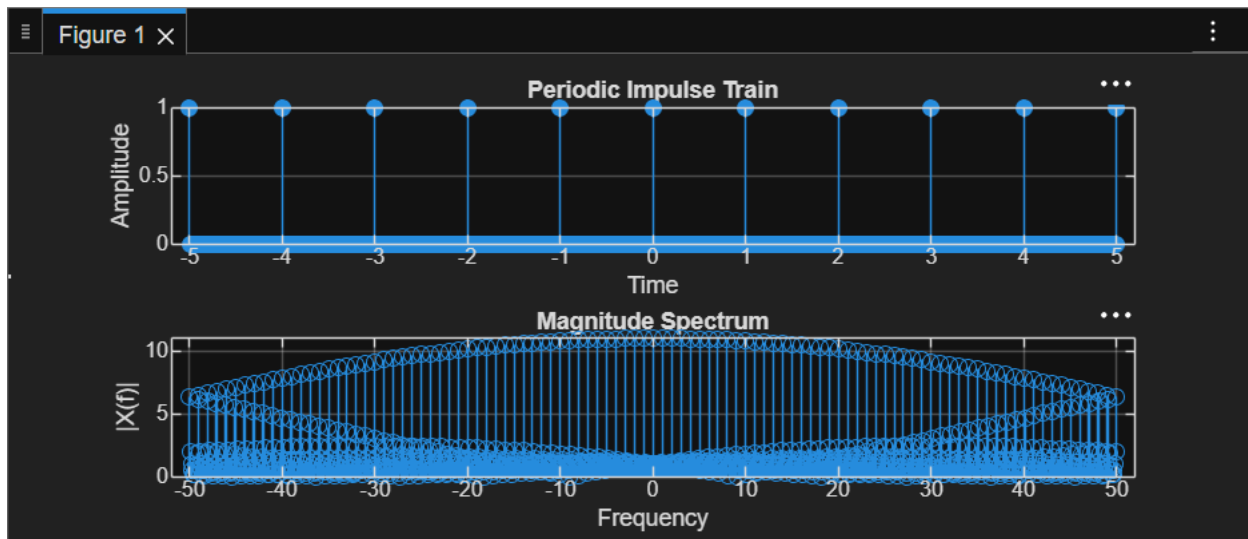
```
x = zeros(size(t));
```

```
x(1:100:end) = 1; % Impulse every 100 samples
```

```
% Fourier Transform
```

```
X = fftshift(fft(x));  
% Frequency axis  
N = length(X);  
f = linspace(-50,50,N);  
% Plotting  
figure;  
subplot(2,1,1);  
stem(t,x,'filled');  
grid on;  
title('Periodic Impulse Train');  
xlabel('Time');  
ylabel('Amplitude');  
subplot(2,1,2);  
stem(f,abs(X));  
grid on;  
title('Magnitude Spectrum');  
xlabel('Frequency');  
ylabel('|X(f)|');
```

Output Waveform:



Result:

The Fourier Transform of a periodic impulse train was obtained using MATLAB. The frequency-domain representation was observed to be another impulse train with discrete spectral lines, confirming the Fourier Transform property of periodic impulse trains and illustrating the relationship between time-domain periodicity and frequency-domain discreteness.

Experiment 8.2

Discrete Time Fourier Transform

Aim:

The Aim of this experiment is to understand and implement the Discrete Time Fourier Transform (DTFT) using MATLAB. The DTFT is a fundamental concept in signal processing, used to analyze the frequency content of discrete-time signals.

Program:

% Discrete-Time Fourier Transform (DTFT)

clc;

clear;

close all;

% Input sequence

x = [1 2 3 4];

% Frequency range

w = -pi:0.01:pi;

% DTFT computation

X = zeros(size(w));

for k = 1:length(w)

for n = 1:length(x)

X(k) = X(k) + x(n)*exp(-1j*w(k)*(n-1));

end

end

% Plotting

figure;

subplot(2,1,1);

stem(0:length(x)-1,x,'filled');

grid on;

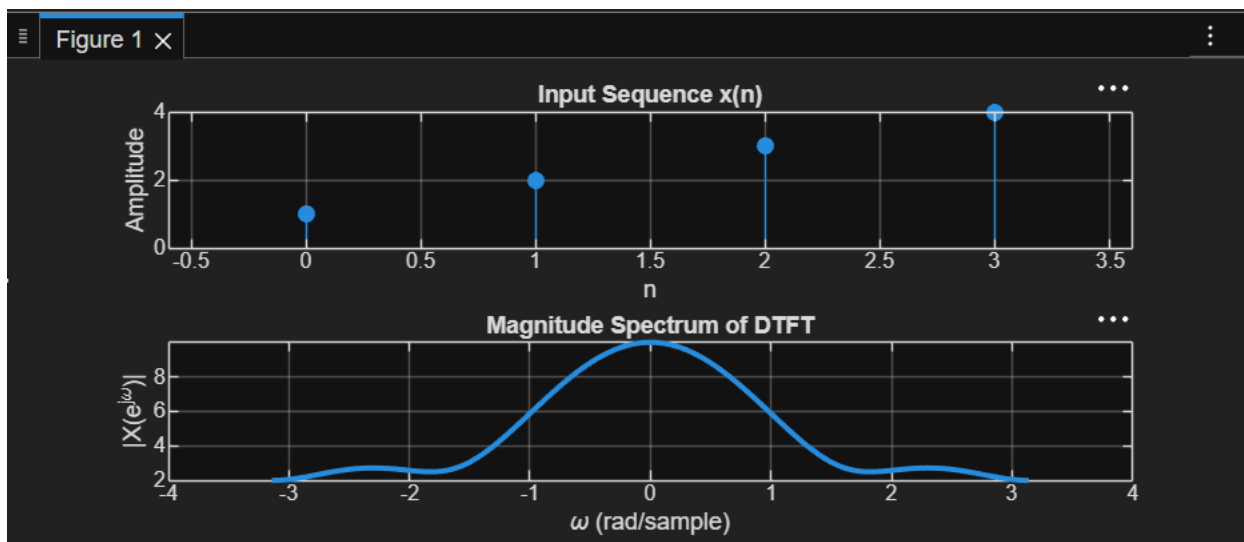
title('Input Sequence x(n)');

xlabel('n');

ylabel('Amplitude');

```
subplot(2,1,2);  
plot(w,abs(X),'LineWidth',2);  
  
grid on;  
title('Magnitude Spectrum of DTFT');  
xlabel('\omega (rad/sample)');  
ylabel('|X(e^{j\omega})|');
```

Output Waveform:



Result:

The Discrete-Time Fourier Transform (DTFT) of the sequence $x(n)=\{1,2,3,4\}$ was computed using MATLAB. The magnitude spectrum was obtained and plotted, illustrating the frequency domain representation of the discrete-time signal.